

## WINCHESTER, VA, USA

WMO: 724053

Lat: 39.143N

Lon: 78.144W

Elev: 727

StdP: 14.31

Time Zone: -5.00 (NAE)

Period: 95-19

WBAN: 03717

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB  |             | Humidification DP/MCDB and HR |            |             |            |            |             | Coldest Month WS/MCDB |             |             |             | MCWS/PCWD to 99.6% DB |            | WSF          |
|---------------|-------------|-------------|-------------------------------|------------|-------------|------------|------------|-------------|-----------------------|-------------|-------------|-------------|-----------------------|------------|--------------|
|               |             |             | 99.6%                         |            |             | 99%        |            |             | 0.4%                  |             | 1%          |             |                       |            |              |
|               | 99.6%       | 99%         | DP                            | HR         | MCDB        | DP         | HR         | MCDB        | WS                    | MCDB        | WS          | MCDB        | MCWS                  | PCWD       |              |
| (a)<br>1      | (b)<br>13.0 | (c)<br>17.9 | (d)<br>-3.5                   | (e)<br>4.7 | (f)<br>18.3 | (g)<br>2.6 | (h)<br>6.5 | (i)<br>21.1 | (j)<br>26.9           | (k)<br>33.5 | (l)<br>24.9 | (m)<br>32.9 | (n)<br>6.9            | (o)<br>300 | (p)<br>0.434 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |             |             |             |             |             | Evaporation WB/MCDB |             |             |             |             |             | MCWS/PCWD to 0.4% DB |            |
|---------------|------------------------|-----------------|-------------|-------------|-------------|-------------|-------------|---------------------|-------------|-------------|-------------|-------------|-------------|----------------------|------------|
|               |                        | 0.4%            |             | 1%          |             | 2%          |             | 0.4%                |             | 1%          |             | 2%          |             |                      |            |
|               |                        | DB              | MCWB        | DB          | MCWB        | DB          | MCWB        | WB                  | MCDB        | WB          | MCDB        | WB          | MCDB        | MCWS                 | PCWD       |
| (a)<br>7      | (b)<br>20.4            | (c)<br>93.1     | (d)<br>74.2 | (e)<br>90.4 | (f)<br>73.5 | (g)<br>88.2 | (h)<br>72.7 | (i)<br>77.2         | (j)<br>88.6 | (k)<br>75.9 | (l)<br>86.3 | (m)<br>74.6 | (n)<br>84.2 | (o)<br>7.8           | (p)<br>280 |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 73.4                            | 127.4 | 81.2 | 72.7 | 124.3 | 80.5 | 72.1 | 121.8 | 79.8 | 41.0          | 88.8 | 39.7 | 85.9 | 38.6 | 84.0 | 83.5              |

## Extreme Annual Design Conditions

| Extreme Annual WS |          |          | Extreme Annual Temperature |          |                    |         | n-Year Return Period Values of Extreme Temperature |           |            |           |            |           |            |           |    |    |
|-------------------|----------|----------|----------------------------|----------|--------------------|---------|----------------------------------------------------|-----------|------------|-----------|------------|-----------|------------|-----------|----|----|
|                   |          |          | Mean                       |          | Standard Deviation |         | n=5 years                                          |           | n=10 years |           | n=20 years |           | n=50 years |           |    |    |
| 1%                | 2.5%     | 5%       | Min                        | Max      | Min                | Max     | Min                                                | Max       | Min        | Max       | Min        | Max       | Min        | Max       | DB | WB |
| (a) 23.2          | (b) 19.3 | (c) 16.7 | (e) 5.5                    | (f) 97.9 | (g) 4.8            | (h) 6.3 | (i) 2.1                                            | (j) 102.4 | (k) -0.7   | (l) 106.1 | (m) -3.4   | (n) 109.6 | (o) -6.8   | (p) 114.2 |    |    |
| (a) 23.2          | (b) 19.3 | (c) 16.7 | (e) 4.3                    | (f) 79.4 | (g) 4.4            | (h) 1.6 | (i) 1.1                                            | (j) 80.6  | (k) -1.5   | (l) 81.5  | (m) -4.0   | (n) 82.4  | (o) -7.2   | (p) 83.6  |    |    |

## Monthly Climatic Design Conditions

|                                                                               |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|-------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                               |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg       | 55.5   | 34.3  | 36.9  | 44.2  | 54.8  | 63.8  | 72.0  | 76.2  | 74.7  | 67.8  | 56.3  | 45.7  | 37.5 |
|                                                                               | DBStd       | 16.87  | 10.33 | 9.61  | 9.70  | 8.75  | 7.93  | 6.16  | 5.29  | 5.12  | 6.73  | 8.22  | 8.62  | 8.53 |
|                                                                               | HDD50       | 1776   | 500   | 380   | 235   | 46    | 2     | 0     | 0     | 0     | 0     | 30    | 183   | 400  |
|                                                                               | HDD65       | 4619   | 950   | 788   | 647   | 325   | 122   | 14    | 1     | 2     | 47    | 291   | 579   | 853  |
|                                                                               | CDD50       | 3766   | 15    | 12    | 56    | 189   | 430   | 660   | 813   | 765   | 534   | 226   | 54    | 12   |
|                                                                               | CDD65       | 1136   | 0     | 0     | 3     | 18    | 85    | 224   | 349   | 302   | 131   | 23    | 1     | 0    |
|                                                                               | CDH74       | 10478  | 0     | 4     | 33    | 226   | 801   | 2019  | 3366  | 2709  | 1112  | 198   | 9     | 1    |
|                                                                               | CDH80       | 3904   | 0     | 0     | 7     | 53    | 244   | 727   | 1428  | 1035  | 369   | 40    | 1     | 0    |
| Wind                                                                          | WSAvg       | 5.6    | 7.0   | 7.0   | 7.3   | 6.9   | 5.2   | 4.9   | 4.4   | 4.0   | 4.1   | 5.1   | 5.8   | 6.0  |
| Precipitation                                                                 | PrecAvg     | 40.6   | 2.7   | 2.2   | 3.7   | 3.2   | 3.7   | 4.0   | 3.6   | 3.3   | 4.4   | 3.1   | 3.1   | 3.0  |
|                                                                               | PrecMax     | 57.8   | 7.1   | 5.8   | 8.2   | 6.3   | 7.7   | 8.0   | 6.7   | 6.0   | 10.4  | 7.6   | 7.3   | 5.3  |
|                                                                               | PrecMin     | 31.9   | 1.0   | 0.5   | 0.0   | 1.4   | 1.1   | 1.1   | 1.0   | 1.3   | 0.2   | 0.3   | 0.4   | 1.1  |
|                                                                               | PrecStd     | 7.0    | 1.6   | 1.3   | 1.7   | 1.3   | 1.8   | 1.9   | 1.5   | 1.3   | 2.9   | 2.0   | 1.7   | 1.2  |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 66.5  | 71.5  | 79.0  | 86.2  | 90.4  | 94.9  | 99.0  | 95.7  | 92.7  | 85.2  | 73.4  | 68.3 |
|                                                                               |             | MCWB   | 58.8  | 57.4  | 60.7  | 64.4  | 72.1  | 74.7  | 75.2  | 75.3  | 71.2  | 69.6  | 60.4  | 59.1 |
|                                                                               | 2%          | DB     | 61.4  | 63.0  | 71.9  | 80.7  | 86.2  | 90.4  | 94.2  | 91.2  | 88.3  | 79.8  | 69.5  | 61.1 |
|                                                                               |             | MCWB   | 55.1  | 52.2  | 57.3  | 62.8  | 70.2  | 73.9  | 74.8  | 74.3  | 71.0  | 66.1  | 58.2  | 53.9 |
|                                                                               | 5%          | DB     | 55.1  | 56.8  | 65.6  | 75.2  | 81.9  | 87.9  | 90.6  | 89.6  | 84.1  | 74.6  | 63.8  | 54.8 |
|                                                                               |             | MCWB   | 49.0  | 47.7  | 54.4  | 59.9  | 67.6  | 73.1  | 74.0  | 73.4  | 69.5  | 63.3  | 56.0  | 48.8 |
|                                                                               | 10%         | DB     | 49.7  | 51.8  | 61.2  | 70.8  | 78.8  | 84.1  | 88.0  | 86.1  | 81.0  | 70.4  | 60.8  | 50.3 |
|                                                                               |             | MCWB   | 44.0  | 44.5  | 51.2  | 58.0  | 65.8  | 71.2  | 73.5  | 72.3  | 67.9  | 62.0  | 52.7  | 44.7 |
| Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 61.0  | 59.0  | 63.2  | 68.9  | 75.0  | 77.6  | 78.9  | 78.8  | 75.5  | 72.0  | 64.6  | 60.8 |
|                                                                               |             | MCDB   | 64.3  | 67.4  | 75.2  | 80.6  | 86.4  | 88.9  | 90.1  | 90.0  | 84.8  | 80.1  | 70.7  | 65.6 |
|                                                                               | 2%          | WB     | 56.4  | 53.7  | 59.7  | 65.0  | 72.3  | 75.9  | 77.4  | 77.0  | 73.6  | 69.1  | 61.3  | 55.3 |
|                                                                               |             | MCDB   | 61.1  | 60.9  | 67.5  | 75.3  | 82.6  | 86.8  | 89.3  | 87.8  | 82.6  | 75.4  | 66.0  | 59.9 |
|                                                                               | 5%          | WB     | 50.2  | 48.6  | 56.3  | 62.4  | 70.2  | 74.3  | 76.0  | 75.4  | 72.0  | 66.2  | 57.6  | 50.0 |
|                                                                               |             | MCDB   | 55.3  | 55.9  | 64.4  | 72.5  | 78.7  | 84.4  | 87.0  | 84.6  | 80.3  | 72.9  | 63.0  | 54.3 |
|                                                                               | 10%         | WB     | 44.2  | 44.7  | 52.3  | 59.7  | 67.9  | 72.8  | 74.6  | 74.0  | 70.3  | 63.2  | 53.8  | 45.4 |
|                                                                               |             | MCDB   | 48.6  | 50.2  | 60.2  | 69.0  | 75.9  | 81.7  | 84.6  | 82.4  | 76.8  | 68.0  | 58.6  | 49.0 |
| Mean Daily<br>Temperature<br>Range                                            | 5% DB       | MDBR   | 15.3  | 16.8  | 19.0  | 21.5  | 20.7  | 20.1  | 20.4  | 20.2  | 20.7  | 20.1  | 18.1  | 15.0 |
|                                                                               |             | MCDBR  | 22.2  | 25.8  | 27.3  | 28.5  | 25.6  | 23.3  | 23.5  | 23.5  | 25.9  | 26.1  | 24.1  | 21.9 |
|                                                                               |             | MCWBR  | 16.9  | 17.7  | 16.6  | 15.2  | 12.7  | 10.0  | 8.7   | 9.1   | 10.9  | 14.2  | 16.2  | 16.7 |
|                                                                               | 5% WB       | MCDBR  | 20.6  | 23.9  | 24.3  | 24.2  | 22.7  | 20.9  | 20.3  | 19.8  | 20.5  | 22.3  | 20.9  | 20.3 |
| MCWBR                                                                         |             | 17.1   | 17.7  | 16.3  | 14.5  | 12.1  | 9.9   | 8.7   | 8.7   | 10.2  | 13.4  | 15.8  | 17.1  |      |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub        | 0.301  | 0.317 | 0.351 | 0.384 | 0.421 | 0.486 | 0.504 | 0.501 | 0.418 | 0.360 | 0.318 | 0.303 |      |
|                                                                               | taud        | 2.473  | 2.406 | 2.380 | 2.317 | 2.236 | 2.085 | 2.059 | 2.046 | 2.262 | 2.423 | 2.504 | 2.499 |      |
|                                                                               | Ebn at Noon | 278    | 287   | 287   | 283   | 273   | 255   | 249   | 246   | 261   | 268   | 269   | 268   |      |
|                                                                               | Edh at Noon | 25     | 30    | 35    | 40    | 44    | 51    | 52    | 51    | 39    | 30    | 24    | 23    |      |
| All-Sky Solar<br>Radiation                                                    | RadAvg      | 630    | 898   | 1199  | 1525  | 1697  | 1880  | 1835  | 1646  | 1352  | 1008  | 724   | 550   |      |
|                                                                               | RadStd      | 56     | 108   | 69    | 124   | 165   | 139   | 105   | 85    | 171   | 115   | 63    | 52    |      |

## Historical Trends

|                         |  | Heating |        | Cooling |       |       | Degree-Days |       |       |       |      |
|-------------------------|--|---------|--------|---------|-------|-------|-------------|-------|-------|-------|------|
|                         |  | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |      |
|                         |  | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   | (m)  |
| Station Only            |  | N/A     | N/A    | N/A     | -1.33 | N/A   | N/A         | N/A   | N/A   | N/A   | N/A  |
| Regional (60 neighbors) |  | +0.74   | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | +223  | +122 |

Nomenclature: See separate page